

Global solutions for the interpolated stochastic heat and wave equation with a super-linear diffusion term

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Abstract

In this paper we prove the existence of a global solution to the interpolated stochastic heat and wave equation driven by a space-time white noise and with a super-linear non-linear diffusion term.

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1 Introduction and main result

In this chapter, we study the interpolated stochastic heat and wave equation (ISHWE) on the whole space $\mathbb{R}^d$,

$$\begin{cases} \left(\partial_t^\beta + \frac{\nu}{2}(-\Delta)^{\alpha/2} \right) u(t, x) = I_t^\gamma \left[\sigma(u(t, x)) \dot{W}(t, x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = u_0(x) & \beta \in (0, 1] \\ u(0, \cdot) = u_0(x), \quad \partial_t u(0, \cdot) = v_0(x) & \beta \in (1, 2), \end{cases} \quad (1.1)$$

E:SupLinSPD

where the fractional differential operators ∂_t^β , $(-\Delta)^{\alpha/2}$ and I_t^γ respectively denote the *Caputo derivative*, the *fractional Laplacian* and the *Riemann-Liouville fractional integral operator*. The noise, $\dot{W}$ is taken to be space-time white, however some comments will be made in Section 4

below about the current progress with a colored noise. The initial data, u_0 and v_0 , are assumed to be Borel measurable functions which are Bounded on $\mathbb{R}^d$. The nonlinearity, σ , is *locally Lipschitz* in the sense that the following asymptotic relation holds as $|x|, |z| \rightarrow \infty$:

$$|\sigma(x) - \sigma(z)| \leq \sigma_2 |x - z| [\ln_+(|x - z|)]^\delta, \quad (1.2)$$

where $\sigma_2, \delta > 0$ and $\ln_+(z) = \ln(z \vee e)$ for $z > 0$. Further, this implies that as $|x| \rightarrow \infty$,

$$|\sigma(x)| \leq \sigma_1 + \sigma_2 |x| [\ln(|x|)]^\delta, \quad (1.3)$$

where $\sigma_1 = \sigma(0)$.

For any $T > 0$ with $t \in [0, T]$ and $x \in \mathbb{R}^d$, the solution to (1.1) is understood in the *mild* sense as the solution to the following stochastic integral equation:

$$u(t, x) = J_0(t, x) + I(t, x),$$

where

$$J_0(t, x) = \begin{cases} [Z(t, \cdot) * u_0](x) & \beta \in (0, 1] \\ [Z(t, \cdot) * v_0](x) + [Z^*(t, \cdot) * u_0](x) & \beta \in (1, 2) \end{cases}$$

$$I(t, x) = \int_0^t \int_{\mathbb{R}^d} Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy).$$

The stochastic integral above is a *Walsh integral* (e.g. see [Wal86]) and the fundamental solution is a triple, $\{Z, Z^*, Y\}$, and each member of the triple is defined through the Fox-H function, however, one can more compactly define them through their Fourier transforms [CHN19, Theorem 4.1]:

$$\begin{aligned} \mathcal{F}Z(t, \cdot)(\xi) &= t^{\lceil \beta \rceil - 1} E_{\beta, \lceil \beta \rceil} \left(-2^{-1} \nu t^\beta |\xi|^\alpha \right), \\ \mathcal{F}Z^*(t, \cdot)(\xi) &= E_{\beta, 1} \left(-2^{-1} \nu t^\beta |\xi|^\alpha \right), \\ \mathcal{F}Y(t, \cdot)(\xi) &= t^{\beta + \gamma - 1} E_{\beta, \beta + \gamma} \left(-2^{-1} \nu t^\beta |\xi|^\alpha \right), \end{aligned}$$

where $E_{a,b}(z)$ denotes the *two-parameter Mittag-Leffler function* [Pod99]:

$$E_{a,b}(z) := \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(ak + b)}, \quad a > 0, b > 0,$$

and Γ is the standard *Gamma-function*.

We mention that under the special case where $\alpha \in (1, 2)$, $\beta \in (0, 1]$ and $\gamma = 1 - \beta$, that $Z = Z^* = Y$, which is easily seen from their Fourier transforms given above. Because of these equalities, we simply let Y denote the fundamental solution under this situation. Moreover, under this case there is a probabilistic representation for Y which is used by the authors in [FLN19; FN17; MN15; FLO17; MN16]. Indeed, let X_t denote a symmetric α -stable process with density $p(t, x)$. Let $D = \{D_r, r \geq 0\}$ denote a β -stable subordinator and E_t its first passage time. Then it is known that the density of the time changed processes X_{E_t} is given by $Y(t, x)$ and we have that

$$Y(t, x) = \int_0^\infty p(s, x) f_{E_t}(s) ds,$$

where

$$f_{E_t}(s) = t^{\beta-1} x^{-1-1/\beta} g_\beta \left(tx^{-1/\beta} \right),$$

where $g_\beta(\cdot)$ is the density function of D_1 and is smooth on the real line with $g_\beta(x) = 0$ for $x \leq 0$.

The goal of this paper is to prove the existence and uniqueness of a global solution to (1.1). The work is highly motivated by the recent work by Millet et al [MS21]. It is a well studied phenomena that either a super-linear drift or diffusion term may cause blow-up of the solution. As for the stochastic heat equation, we direct the reader to [FN21; MS93; DKZ19; BG09]. On the other hand, the only other work that we are aware of that is dedicated to proving some non-existence results of (1.1) is [AMN20]. So, to the best of our knowledge, this is the first work on proving the global existence of a solution to (1.1) with a super-linear diffusion term (see Theorem 1.1 below). However, as we will see, the calculations given below are essentially identical to those performed in [MS21] which is the cause for our motivation.

Here is the main result of this chapter. The proof will be postponed until Section 3 below.

C:MainSupLin

Theorem 1.1. *Suppose that σ satisfies (1.2) and the initial data are Borel measurable functions. Then for any $R > 0$ and $(t, x) \in [0, T] \times \bar{B}_R(0)$, where $\bar{B}_R(0)$ denotes the closed ball centered around the origin, there exists a random field solution to (1.1) denoted as $(u(t, x) : (t, x) \in [0, T] \times \bar{B}_R(0))$. This solution is unique and satisfies*

$$\sup_{t \in [0, T], |x| \leq R} |u(t, x)| < \infty, \quad \text{almost surely.} \quad (1.4)$$

The proof of Theorem 1.1 uses a standard stopping time argument along with a series of truncations of the super-linear term, σ , in order to construct a solution of (1.1). Indeed, we define

$$\sigma_N(x) = \begin{cases} \sigma(x) & |x| \leq N \\ \text{continuous and constant} & |x| > N, \end{cases}$$

and then associate for each $N > 0$, the corresponding global solution $u_N(t, x)$, whose existence follows due to Theorem 2.1 below. We then define the stopping time τ_N as follows:

$$\tau_N := \inf \left\{ t > 0 : \sup_{|x| \leq R} |u_N(t, x)| \geq N \right\} \wedge T.$$

We prove that $\{t < \tau_N\} \uparrow \Omega$ and then show that the solution to (1.1) exists pathwise for $\omega \in \Omega$. We remind the reader that in the case of a globally Lipschitz diffusion term, one usually proves that the solution exists as a limit in $L^2(\Omega)$. Because of the stopping-time argument, existence in $L^2(\Omega)$ is no longer guaranteed. Moreover, because $L^2(\Omega)$ existence is no longer guaranteed, then the solution may no longer satisfy the standard Ito isometry.

2 Preliminary results for a globally Lipschitz drift term

In this section, we assume that the coefficient σ is globally Lipschitz and therefore satisfies the following inequality:

$$|\sigma(x)| \leq c(\sigma) + L(\sigma)|x|, \quad (2.1)$$

E:SupLinGlo

where $c(\sigma) = \sigma(0)$ and $L(\sigma)$ is the Lipschitz constant of σ and we assume that $L(\sigma) > 0$. The equation (1.1) under the assumption that σ satisfies (2.1) has been studied in [CHN19] where they establish in Theorem 4.1 *ibid* the existence and uniqueness of global random field solutions. In order to do so, one needs to show *Dalang's condition*:

$$\int_0^t ds \int_{\mathbb{R}^d} dy |Y(s, y)|^2 < \infty, \quad \text{for all } t > 0, \quad (2.2)$$

C:Dalang

which is equivalent to (e.g. see Lemma 5.3 *ibid*)

$$d < 2\alpha + \frac{\alpha}{\beta} \min\{2\gamma - 1, 0\} =: \Theta, \quad (2.3)$$

E:DalangEqu

which is further equivalent to

$$\rho(d) > 0 \quad \text{and} \quad d < 2\alpha, \quad (2.4) \quad \boxed{\text{E: DalangEqu}}$$

where

$$\rho(x) := 2\beta + 2\gamma - 1 - \beta x/\alpha. \quad (2.5) \quad \boxed{\text{D: rho}}$$

We now state the existence and uniqueness result proven in [CHN19].

$\boxed{\text{1and2normYZ}}$

Theorem 2.1. [CHN19, Theorem 4.1] Under (2.2), the SPDE (1.1) has a unique (in the sense of versions) random field solution $\{u(t, x) : (t, x) \in (0, \infty) \times \mathbb{R}^d\}$ if the initial data are such that

$$\widehat{C}_T := \sup_{t \in [0, T], x \in \mathbb{R}^d} |J_0(t, x)| < \infty. \quad (2.6) \quad \boxed{\text{E: HomogBdd}}$$

We now state a lemma proven in [CHN19] that calculates the $L^1(\mathbb{R}^d)$ and $L^2(\mathbb{R}^d)$ norms of the fundamental solutions. The lemma will be needed below in some of our calculations.

$\boxed{\text{1and2normYZ}}$

Lemma 2.2. [CHN19, Theorem 4.1 and Lemma 5.5] Assume that $d < 2\alpha$, $\beta \in (0, 2)$, and $\gamma \geq 0$. Then

$$\int_{\mathbb{R}^d} Y(t, x) dx = t^{\beta+\gamma-1} \quad \text{and} \quad \int_{\mathbb{R}^d} Z(t, x) dx = t^{\lceil \beta \rceil - 1}$$

and

$$\int_{\mathbb{R}^d} Y^2(t, x) dx = C_{\#} t^{2(\beta+\gamma-1)-d\beta/\alpha} = C_{\#} t^{\rho-1},$$

for all $t > 0$, where $\rho = \rho(d)$ and

$$C_{\#} := \frac{2}{\Gamma(d/2)(4\pi)^{d/2}} \int_0^\infty u^{d-1} E_{\beta, \beta+\gamma}^2(-u^\alpha) du.$$

Now further assume that $\beta \in (1, 2)$, then

$$\int_{\mathbb{R}^d} Z^*(t, x) dx = 1.$$

Remark 2.3. When $\alpha = \beta = 2$, $\gamma = 0$ and $d = 1$ then Lemma 2.2 gives us that $\int_{\mathbb{R}} Y^2(t, x) dx = t/2$ which coincides with the $L^2(\mathbb{R})$ norm of the wave kernel.

2.1 Some moment bounds

In this section, we will prove some moment bounds and a continuity result for the solution with a globally Lipschitz diffusion term. In the proof of the next proposition, we will use the following two facts:

$$\sup_{t \geq 0} t^k e^{-at} = k^k (ea)^{-k} \text{ for } k \in \mathbb{N} \text{ and } \sup_{t \geq 0} \int_0^t s e^{-as} ds = a^{-2} \text{ for } a > 0. \quad (2.7) \quad \boxed{\text{E: SUPidenti}}$$

$\boxed{\text{P:N(u)Bdd}}$

Proposition 2.4. [MS21, Proposition 3.2] Let u_0 and v_0 be Borel functions satisfying $\|u_0\|_\infty + \|v_0\|_\infty < \infty$. Suppose that $d < 2\alpha$ and $\rho = \rho(d) > 0$, where $\rho(x)$ is defined in (2.5). Then there exists a universal constant $K := K(\alpha, \beta, \gamma, d) > 0$ such that for any

$$a > (8L^2(\sigma)K^2)^{1/\rho}.$$

and

$$p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2} \right].$$

we have that

$$N_{a,p}(u) \leq 2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \quad (2.8) \quad \boxed{\text{E:Nbdd}}$$

where

$$\mathcal{T}_0 = \begin{cases} (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|u_0\|_\infty & \beta \in (0, 1] \\ (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases}$$

Moreover for any $T > 0$,

$$\sup_{x \in \mathbb{R}^d} \mathbb{E}(|u(t, x)|^p) \leq \exp(apt) \left[2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \right]^p, \quad t \in [0, T]. \quad (2.9) \quad \boxed{\text{E:Pnormtbdd}}$$

Proof. First we fix an $a > 0$ and $p \in [2, \infty)$. Using Lemma 2.2 we see that,

$$\begin{aligned} |J(t, x)| &= \begin{cases} |(Z(t, \cdot) * u_0)(x)| & \beta \in (0, 1] \\ |(Z(t, \cdot) * v_0)(x) + (Z^*(t, \cdot) * u_0)(x)| & \beta \in (1, 2) \end{cases} \\ &\leq \begin{cases} t^{\lceil \beta \rceil - 1} \|u_0\|_\infty & \beta \in (0, 1] \\ t^{\lceil \beta \rceil - 1} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2) \end{cases} \end{aligned}$$

Now using (2.7), we see that

$$\begin{aligned} N_{a,p}(J) &\leq \begin{cases} \sup_{t \geq 0} \sup_{x \in \mathbb{R}} t^{\lceil \beta \rceil - 1} \|u_0\|_\infty e^{-at} & \beta \in (0, 1] \\ \sup_{t \geq 0} \sup_{x \in \mathbb{R}} (t^{\lceil \beta \rceil - 1} \|v_0\|_\infty + \|u_0\|_\infty) e^{-at} & \beta \in (1, 2) \end{cases} \\ &= \begin{cases} (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|u_0\|_\infty & \beta \in (0, 1] \\ (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases} \quad (2.10) \quad \boxed{\text{E:Jbdd}} \end{aligned}$$

Now we find an upper bound for $N_{a,p}(I)$. By simultaneously applying the Burkholder-Davies-Gundy inequality and triangle inequality, we get that

$$\begin{aligned} \|I(t, x)\|_p^2 &\leq 4p \left\| \int_0^t ds \int_{\mathbb{R}^d} dy Y^2(t - s, x - y) \sigma^2(u(s, y)) \right\|_{p/2} \\ &\leq 4p \int_0^t ds \int_{\mathbb{R}^d} dy Y^2(t - s, x - y) \|\sigma^2(u(s, y))\|_{p/2}. \end{aligned}$$

Now by applying Lemma 2.2 and the Lipschitz property of σ we get that

$$\|I(t, x)\|_p^2 \leq 8pc^2(\sigma)C_\# \int_0^t (t - s)^{\rho - 1} ds + 8pL^2(\sigma)N_{a,p}^2(u) \int_0^t ds \int_{\mathbb{R}^d} dy e^{2as} Y^2(t - s, x - y).$$

Now under Dalang's condition (2.4), the first integral above can be calculated as

$$\int_0^t (t - s)^{\rho - 1} ds = \frac{t^\rho}{\rho}.$$

By applying Lemma 2.2, we can calculate the second integral as

$$\int_0^t ds \int_{\mathbb{R}} dy e^{2as} Y^2(t - s, x - y) = C_\# \int_0^t ds e^{2as} (t - s)^{\rho - 1}.$$

Putting this together gives us that

$$\|I(t, x)\|_p^2 \leq 8pc^2(\sigma)C_\# \frac{t^\rho}{\rho} + 8pL^2(\sigma)N_{a,p}^2(u)C_\# \int_0^t ds e^{2as} (t - s)^{\rho - 1}.$$

Now by multiplying both sides by e^{-2at} , we see that

$$\begin{aligned} \|I(t, x)\|_p^2 e^{-2at} &\leq \frac{8pc^2(\sigma)C_{\#}}{\rho} t^{\rho} e^{-2at} \\ &\quad + 8pL^2(\sigma)N_{a,p}^2(u)C_{\#} \int_0^t ds e^{-2a(t-s)}(t-s)^{\rho-1}. \end{aligned}$$

We now do a change of variables and again recall the definition of the Gamma function to see that

$$\begin{aligned} \|I(t, x)\|_p^2 e^{-2at} &\leq \frac{8pc^2(\sigma)C_{\#}}{\rho} t^{\rho} e^{-2at} + 8pL^2(\sigma)N_{a,p}^2(u)C_{\#} \int_0^{\infty} du e^{-2au}(u)^{\rho-1} \\ &= \frac{8pc^2(\sigma)C_{\#}}{\rho} t^{\rho} e^{-2at} + 8pL^2(\sigma)N_{a,p}^2(u) \frac{C_{\#}\Gamma(\rho)}{(2a)^{\rho}} \end{aligned}$$

where the last line is true if Dalang's condition [CHN19] holds. If we now apply (2.7) and also recall that $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$, then we see that

$$N_{a,p}(I) \leq c(\sigma) \sqrt{\frac{8pC_{\#}}{\rho}} \left(\frac{\rho}{2ea}\right)^{\rho/2} + L(\sigma)N_{a,p}(u) \sqrt{\frac{8pC_{\#}\Gamma(\rho)}{(2a)^{\rho}}}. \quad (2.11) \quad \boxed{\text{E:SoleMil3.}}$$

To simplify things, we now let

$$K := \max \left\{ \sqrt{\frac{8C_{\#}}{\rho}} \left(\frac{\rho}{2e}\right)^{\rho/2}, \sqrt{\frac{8C_{\#}\Gamma(\rho)}{2^{\rho}}} \right\}$$

then,

$$N_{a,p}(I) \leq \frac{K\sqrt{p}}{a^{\rho/2}} (c(\sigma) + L(\sigma)N_{a,p}(u)). \quad (2.12) \quad \boxed{\text{E:Ibdd}}$$

Now by combining (2.10) and (2.12), we see that

$$N_{a,p}(u) \leq \begin{cases} (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|u_0\|_{\infty} & \beta \in (0, 1] \\ (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|v_0\|_{\infty} + \|u_0\|_{\infty} & \beta \in (1, 2). \end{cases} \quad (2.13)$$

$$+ K \frac{\sqrt{p}c(\sigma)}{a^{\rho/2}} + K \frac{\sqrt{p}L(\sigma)}{a^{\rho/2}} N_{a,p}(u). \quad (2.14) \quad \boxed{\text{E:Nbddgen}}$$

It is important to note that if we now restrict ourselves to the case of $\alpha = \beta = 2$, $d = 1$ and $\gamma = 0$, then (2.11) and (2.14) recover [MS21, (3.9) and (3.10)] respectively. Now choose $a > 0$ such that

$$a > (8L^2(\sigma)K^2)^{1/\rho}.$$

This implies that the following interval is nonempty:

$$\left[2, \frac{a^{\rho}}{4L^2(\sigma)K^2} \right].$$

In addition, for any p in this interval, we have that

$$\frac{\sqrt{p}L(\sigma)K}{a^{\rho/2}} \leq \frac{1}{2},$$

and we also have

$$\frac{\sqrt{p}}{a^{\rho/2}} \leq \frac{1}{2KL(\sigma)}.$$

This along with (2.14) implies that

$$N_{a,p}(u) \leq 2 \begin{cases} ([\beta] - 1)^{[\beta]-1} (ea)^{1-[\beta]} \|u_0\|_\infty & \beta \in (0, 1] \\ ([\beta] - 1)^{[\beta]-1} (ea)^{1-[\beta]} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2) \end{cases} + \frac{c(\sigma)}{L(\sigma)}$$

and this is precisely (2.8). Note that (2.9) follows immediately from (2.8). $\square$

We will need to recall the following result proved in [CHN19] in order to prove Proposition 2.7 below.

P:YInc

Proposition 2.5. [CH21, Proposition 2.1 and 2.2] *Suppose that $\alpha \in (0, 2]$, $\beta \in (0, 2)$, $\gamma > 0$, and (2.3) holds. Consider $\rho = \rho(d)$ which is defined above in (2.5). Then $Y(t, x) = Y_{\alpha, \beta, \gamma, d}(t, x)$ satisfies the following:*

P:YIncCase1

1. *Suppose that $0 < \theta < (\Theta - d) \wedge 2$ where Θ is given in (2.3), $0 < t, r \leq T$ for some $T > 0$, $\beta \leq 1$ and $\gamma \leq [\beta] - \beta$. Then there exists some $C := C(\alpha, \beta, \gamma, \nu, d, \theta, T)$ such that*

$$\int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \leq C \left(|t-r|^\rho + |x-z|^\theta \right).$$

P:YIncCase2

2. *If we only assume that $\alpha \in (0, 2]$, $\beta \in (0, 2)$, $\gamma > 0$, $0 < t, r \leq T$ for some $T > 0$ and (2.3) holds then for $0 < \theta < (\Theta - d) \wedge 2$ there exists some $C := C(\alpha, \beta, \gamma, \nu, d, \theta, T)$ such that*

$$\int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \leq C \left(|t-r|^q + |x-z|^\theta \right),$$

where

$$0 < q < \rho.$$

Remark 2.6. In Proposition 2.5 above, we may often ignore the dependence of C on $\alpha, \beta, \gamma, \nu, d$ and just write $C(\theta, T)$.

P:NormInc

Proposition 2.7. *Let $a > 0$ and suppose that $0 < \theta < (\Theta - d) \wedge 2$ and $\rho = \rho(d) > 0$. Then for any $x, z \in \mathbb{R}$, $p \geq 2$ and for any $T > 0$ with $t \in [0, T]$, the stochastic integral, $I(t, x)$, satisfies the following:*

$$\frac{\|I(t, x) - I(r, z)\|_p}{(|t-r|^q + |x-z|^\theta)^{1/2}} \leq C(p, \theta, T) [\mathcal{M}_1 + \mathcal{M}_2 e^{aT} N_{a,p}(u)], \quad (2.15)$$

E:NormInc_N

where

$$\mathcal{M}_1 = \sqrt{p}c(\sigma) \quad \text{and} \quad \mathcal{M}_2 = \sqrt{p}L(\sigma),$$

and

1. $0 < q \leq \rho$ under Case 1 of Proposition 2.5,
2. $0 < q < \rho$ under Case 2 of Proposition 2.5.

Moreover, for any

$$a > (8L^2(\sigma)K^2)^{1/\rho}.$$

and

$$p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2} \right]$$

we have that

$$\frac{\|I(t, x) - I(r, z)\|_p}{(|t-r|^q + |x-z|^\theta)^{1/2}} \leq C(p, \theta, T) \left[\mathcal{M}_1 + \mathcal{M}_2 e^{aT} \left(2 + \frac{c(\sigma)}{L(\sigma)} \right) \right]. \quad (2.16)$$

E:NormInc_c

Proof. Note that because of the initial conditions, we see that $\|u(t, x) - u(r, z)\|_p = \|I(t, x) - I(r, z)\|_p$. We now apply Burkholder-Davies-Gundy inequality along with the triangle inequality, as was done in the proof of Proposition 2.4, to see that

$$\begin{aligned} \|I(t, x) - I(r, z)\|_p^2 &\leq 4p \left\| \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \sigma^2(u(s, y)) \right\|_{p/2} \\ &\leq 4p \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \|\sigma^2(u(s, y))\|_{p/2}. \end{aligned}$$

Now by applying the Lipchitz condition on σ gives us that

$$\begin{aligned} \|I(t, x) - I(r, z)\|_p^2 &\leq 8p \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \left(c^2(\sigma) + L^2(\sigma) \|u(s, y)\|_p^2 \right) \\ &\leq 8p \left[c^2(\sigma) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \right. \\ &\quad \left. + L^2(\sigma) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \exp(2as) N_{a,p}^2(u) \right] \\ &\leq 8p \left[c^2(\sigma) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \right. \\ &\quad \left. + L^2(\sigma) \exp(2aT) N_{a,p}^2(u) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \right]. \end{aligned}$$

We now apply Proposition 2.5 to see that

$$\begin{aligned} \|I(t, x) - I(r, z)\|_p^2 &\leq 8p \left[c^2(\sigma) C(\theta, T) \left(|t-r|^q + |x-z|^\theta \right) \right. \\ &\quad \left. + L^2(\sigma) \exp(2aT) N_{a,p}^2(u) C(\theta, T) \left(|t-r|^q + |x-z|^\theta \right) \right], \end{aligned}$$

where $C(\theta, T)$ is as in Proposition 2.5. Now by taking the square root of both sides and applying the identity $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$ yields

$$\|I(t, x) - I(r, z)\|_p \leq 2\sqrt{2C(\theta, T)p} \left[c(\sigma) + L(\sigma) \exp(aT) N_{a,p}(u) \right] \left(|t-r|^q + |x-z|^\theta \right)^{1/2}.$$

This proves (2.15). It is clear that (2.16) follows directly from (2.15) and Proposition 2.4. $\square$

P:Holder

Proposition 2.8. *The stochastic integral, I , has a version, still denoted by I that is η_1 -Hölder continuous in time and η_2 -Hölder continuous in space with $0 < \eta_1 < q/2$ and $0 < \eta_2 < \theta/2$ where $0 < \theta < (\Theta - d) \wedge 2$ and*

1. $0 < q \leq \rho$ under Case 1 of Proposition 2.5,
2. $0 < q < \rho$ under Case 2 of Proposition 2.5.

Moreover, if we consider

$$a > (8L^2(\sigma)K^2)^{1/\rho} \quad \text{and} \quad p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2} \right],$$

where K is the constant from Proposition 2.4, then

$$\mathbb{E} \left[\sup_{t \in [0, T], |x| \leq R} |u(t, x)|^p \right] \leq 2^{p-1} \widehat{C}_T^p + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2^p e^{apT} \left(2 + \frac{c(\sigma)}{L(\sigma)} \right)^p \right],$$

where $\widehat{C}_T$ is defined in (2.6).

Proof. Proposition 2.7 implies that

$$\|I(t, x) - I(r, z)\|_p \leq C(p, \theta, T) [\mathcal{M}_1 + \mathcal{M}_2 e^{aT} N_{a,p}(u)] \left(|t - r|^q + |x - z|^\theta \right)^{1/2}. \quad (2.17)$$

By Kolmogorov's continuity theorem, there exists a version of I , which we denote by I , that is η_1 -Hölder continuous in time and η_2 -Hölder continuous in space with $0 < \eta_1 < q/2$ and $0 < \eta_2 < \theta/2$.

Next, note that by triangle inequality,

$$\begin{aligned} |u(t, x)|^p &\leq 2^{p-1} (\|J_0(\cdot, \circ)\|_\infty^p + |u(t, x) - J_0(t, x)|^p) \\ &\leq 2^{p-1} \left(\widehat{C}_T^p + \sup_{t \in [0, T], |x| \leq R} |I(t, x)|^p \right) \\ &= 2^{p-1} \left(\widehat{C}_T^p + \sup_{t \in [0, T], |x| \leq R} |I(t, x)|^p \right), \end{aligned}$$

where we recall that $\widehat{C}_T$ defined in (2.6). Due to the continuity of I on the compact set $[0, T] \times \bar{B}_R(0)$, there exists a point $(t_0, x_0) \in [0, T] \times \bar{B}_R(0)$ such that

$$\sup_{t \in [0, T], |x| \leq R} |I(t, x)|^p = |I(t_0, x_0)|^p.$$

Thus, for any $(t, x) \in [0, T] \times \bar{B}_R(0)$,

$$|u(t, x)|^p \leq 2^{p-1} \left(\widehat{C}_T^p + |I(t_0, x_0)|^p \right)$$

which in return implies that

$$\sup_{t \in [0, T], |x| \leq R} |u(t, x)|^p \leq 2^{p-1} \left(\widehat{C}_T^p + |I(t_0, x_0)|^p \right).$$

Now by taking the expectation of both sides and applying Proposition 2.7 gives us

$$\begin{aligned} \mathbb{E} \left[\sup_{t \in [0, T], |x| \leq R} |u(t, x)|^p \right] &\leq 2^{p-1} \widehat{C}_T^p + 2^{p-1} C(p, \theta, T)^p \left((2T)^{q/2} + (2R)^{\theta/2} \right)^p [\mathcal{M}_1 + \mathcal{M}_2 e^{aT} N_{a,p}(u)]^p \\ &= 2^{p-1} \widehat{C}_T^p + C(p, \theta, T, R) [\mathcal{M}_1^p + \mathcal{M}_2^p e^{apT} N_{a,p}(u)^p]. \end{aligned}$$

Now if we consider

$$a > (8L^2(\sigma)K^2)^{1/\rho} \quad \text{and} \quad p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2} \right],$$

then by Proposition 2.4, we have that

$$\mathbb{E} \left[\sup_{t \in [0, T], |x| \leq R} |u(t, x)|^p \right] \leq 2^{p-1} \widehat{C}_T^p + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2^p e^{apT} \left(2 + \frac{c(\sigma)}{L(\sigma)} \right)^p \right].$$

□

Remark 2.9. The Hölder continuity in time proven above in Proposition 2.8 recovers the result proven by Foondun and Nane in [FN17, Theorem 1.9] if we restrict ourself to the case where $\beta \in (0, 1)$ and $\gamma = 1 - \beta$.

3 Proof of Theorem 1.1

Proof. We first solve a *truncated* version of (1.1). By this, we mean (1.1) with σ replaced by σ_N where in the case of $d = 1$,

$$\sigma_N(x) = \sigma(x)\mathbf{1}_{\{|x| \leq N\}} + \sigma(N)\mathbf{1}_{\{x > N\}} + \sigma(-N)\mathbf{1}_{\{x \leq -N\}},$$

and in higher dimensions σ_N is defined similarly. Note that since σ satisfies (1.2), then σ_N is Lipschitz continuous with Lipschitz constant $L(\sigma_N) := \sigma_2 \ln(2N)^\delta$. In other words,

$$|\sigma_N(x)| \leq \sigma_1 + \sigma_2 \ln(2N)^\delta |x|.$$

Hence we can apply Theorem 2.1 and see that there exists an unique solution of the truncated version of (1.1), which we denote by $u_N := \{u_N(t, x) : (t, x) \in [0, T] \times \mathbb{R}\}$. In addition, Proposition 2.8 implies that the solution has a version, still denoted by u_N , which is η_1 -Hölder continuous in time and η_2 -Hölder continuous in space with $0 < \eta_1 < q/2$ and $0 < \eta_2 < \theta/2$.

We will now apply Proposition 2.8 to find an upper bound on the p -norm of u_N . For this, consider

$$a > (8L^2(\sigma_N)K^2)^{1/\rho} \quad \text{and} \quad p \in \left[2, \frac{a^\rho}{4L^2(\sigma_N)K^2}\right].$$

Then Proposition 2.8 implies that

$$\mathbb{E} \left(\sup_{t \in [0, T], x \in [-R, R]} |u_N(t, x)|^p \right) \leq 2^{p-1} + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2^p(N) \exp(apT) \left(2 + \frac{\sigma_1}{L(\sigma_N)} \right) \right] \quad (3.1)$$

E:u_NMomBdd

where we recall that

$$\mathcal{M}_1 = \sqrt{p}\sigma_1 \quad \text{and} \quad \mathcal{M}_2(N) = \sqrt{pL(\sigma_N)} = \sqrt{p\sigma_2 \ln(2N)^\delta}. \quad (3.2)$$

D:M1&2forSi

We now use the above to prove the existence and uniqueness of (1.1) with superlinear σ satisfying (1.2). For any $T > 0$ and $N \geq 2$ with $N \in \mathbb{N}$, we define the following stopping time:

$$\tau_N := \inf \left\{ t > 0 : \sup_{|x| \leq R} |u_N(t, x)| \geq N \right\} \wedge T. \quad (3.3)$$

The uniqueness of the solution and the local property of the stochastic integral imply that on $\{t < \tau_N\}$, $u_N(t, x) = u_{N+k}(t, x)$ for any $k \in \mathbb{N}$. Hence, $(\tau_N)_{N \geq 2}$ is increasing and bounded above by T .

We now momentarily assume that $\sup_N \tau_N = T$ which would imply that $\{t < \tau_N\} \uparrow \Omega$. On each $\{t < \tau_N\}$, define $(u(t, x) : (t, x) \in [0, T] \times \mathbb{R})$ by $u(t, x) = u_N(t, x)$ and hence $u(t, x) = u_{N+k}(t, x)$ for any $k \in \mathbb{N}$. This implies that on $\{t < \tau_N\}$

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy Y(t-s, x-y) \sigma_N(u(s, y)) W(ds, dy).$$

However, on $\{t < \tau_N\}$, we have that $\sigma_N(u_N(t, x)) = \sigma(u(t, x))$ and so u satisfies

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy), \quad \{t < \tau_N\}.$$

Lastly, since $\{t < \tau_N\} \uparrow \Omega$, we conclude that

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy), \quad (t, x) \in [0, T] \times \mathbb{R}. \quad (3.4)$$

E:SupLinMil

It is mentioned that the stochastic integrals are no longer defined in $L^2(\Omega)$. The extension of the integral is found in Dalang and Sanz-Sole's textbook that is not yet published.

We now focus on proving that $\sup_N \tau_N = T$, or equivalently that $\mathbb{P}(\tau_N < T) \rightarrow 0$ as $N \rightarrow \infty$. To do this, note that by Chebychev's inequality

$$\begin{aligned} \mathbb{P}(\tau_N < T) &\leq \mathbb{P}\left(\sup_{t \in [0, T], |x| \leq R} |u_N(t, x)| \geq N\right) \\ &\leq N^{-p} \mathbb{E}\left(\sup_{t \in [0, T], |x| \leq R} |u_N(t, x)|^p\right). \end{aligned}$$

An application of (3.1) gives

$$\begin{aligned} N^{-p} \mathbb{E}\left(\sup_{t \in [0, T], |x| \leq R} |u_N(t, x)|^p\right) &\leq N^{-p} \left(2^{p-1} + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2(N)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{L(\sigma_N)}\right)^p\right]\right) \\ &= \frac{2^{p-1} + C(p, \theta, T, R) \mathcal{M}_1^p}{N^p} + \frac{C(p, T, R) \mathcal{M}_2(N)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{L(\sigma_N)}\right)^p}{N^p}. \end{aligned}$$

The first term clearly converges to 0 as $N \rightarrow \infty$. As for the second term, we expand using (3.2) to see that

$$\begin{aligned} \frac{\mathcal{M}_2(N)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{L(\sigma_N)}\right)^p}{N^p} &= \frac{\left(\sqrt{p\sigma_2 \ln(2N)^\delta}\right)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{\sqrt{p\sigma_2 \ln(2N)^\delta}}\right)^p}{N^p} \\ &\sim \frac{\ln(2N)^{p\delta/2}}{N^p} \rightarrow 0, \end{aligned}$$

where above we use the symbol $\sim$ to mean asymptotically equivalent as $N \rightarrow \infty$. This completes the proof of the Theorem. $\square$

4 Some comments and current progress on a spatially colored noise

colorednoise

Consider equation 1.1 driven by a noise that is white in time and colored space. In other words, W is a 0-mean Gaussian process indexed on the space of Schwartz functions with the following covariance structure:

$$C(\phi, \psi) := \int_0^\infty dt \int_{\mathbb{R}^d} \Gamma(dx) (\phi(t, \cdot) * \tilde{\psi}(t, \cdot))(x), \quad \tilde{\psi}(x) = \psi(-x). \quad (4.1) \quad \text{E:Colored}$$

As per usual, the tempered measure Γ is non-negative and nonnegative definite and when Γ has density f , which will always be assumed for the remainder of this chapter (see Assumption 4.4 below), then we may rewrite (4.1) as the following:

$$C(\phi, \psi) = \int_0^\infty dt \int_{\mathbb{R}^{2d}} dx dy \phi(t, x) f(x - y) \psi(t, y)$$

and by applying Placherel's theorem, this further implies that

$$\int_0^\infty dt \int_{\mathbb{R}^{2d}} dx dy \phi(x) f(x - y) \psi(y) = (2\pi)^{-d} \int_0^\infty dt \int_{\mathbb{R}^{2d}} \mu(d\zeta) \mathcal{F}(\phi(t, \cdot))(\zeta) \overline{\mathcal{F}(\psi(t, \cdot))(\zeta)}.$$

We now impose some assumptions on Γ :

NoiseAssum1

Assumption 4.1.

1. Γ is absolutely continuous with respect to the Lebesgue measure and therefore has a density, $\Gamma(dx) = f(x)dx$. Furthermore, when this is the case, we may often write $\mu(d\zeta) = \varphi(\zeta)d\zeta$ where we have the relation $\varphi = \mathcal{F}f$,
2. the spectral measure μ is such that $\int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1+|\zeta|^\alpha)^2} < \infty$,
3. the spectral density, φ , satisfies the following scaling property: $\varphi(c\xi) = c^{-(d-\lambda)}\varphi(\xi)$ for some $\lambda > 0$.

Remark 4.2. To the best of our knowledge, the only correlation function and spectral density that satisfies the conditions of Assumption 4.1, specifically the third point, is the following fractional Riesz-type kernel:

$$f(x) = \prod_{i=1}^k \frac{1}{|x_{(i)}|^{\Lambda_i}} \quad \text{and} \quad \varphi(\xi) = \prod_{i=1}^k C_{\Lambda_i, d_i} |\xi_{(i)}|^{-(d_i - \Lambda_i)}. \quad (4.2)$$

E:RieszFrac

In (4.2), $k \in \{1, \dots, d\}$ and we partition the d -coordinates of $x = (x_1, \dots, x_d)$ into k distinct groups of size d_i so that $d_1 + \dots + d_k = d$ and $x_{(i)} = (x_{i_1}, \dots, x_{i_{d_i}})$ denotes the coordinates in the i^{th} partition. Moreover, we take $0 < \Lambda_i < d_i$ and denote $\Lambda := \sum_{i=1}^k \Lambda_i$. Lastly, C_{Λ_i, d_i} can be calculated as follows (see [CE22, Example 1.5]):

$$C_{\Lambda, d} = \pi^{-d/2} 2^{-\Lambda} \frac{\Gamma((d - \Lambda)/2)}{\Gamma(\Lambda/2)}.$$

.

Example 4.3. We have the following special cases of (4.2):

1. Setting $k = 1$ in (4.2) recovers the Riesz kernel case. In this case,

$$f(x) = |x|^{-\alpha} \quad \text{and} \quad \varphi(x) = C_{\Lambda, d} |x|^{-(d - \Lambda)}, \quad (4.3)$$

E:RieszNoise

where $0 < \Lambda < d$.

2. Setting $k = d$ in (4.2) recovers the following fractional noise:

$$f(x) = \prod_{i=1}^d |x_i|^{-\Lambda_i} \quad \text{and} \quad \varphi(\xi) = \prod_{i=1}^d C_{\Lambda_i, 1} |\xi_i|^{-(1 - \Lambda_i)} \quad (4.4)$$

E:fracnoise

The SHE with such noise was earlier studied by Hu [Hu01].

Assumption 4.4. Because of the above remark, we will assume that the correlation function is given by the Riesz kernel as in (4.2). Taking into consideration the second point from Assumption 4.1, we must assume that

$$0 < d - \Lambda < d \quad \text{and} \quad d - \Lambda + 2\alpha > d.$$

However, as we will see in Theorem 4.6 below, we will also need to assume that

$$\lambda\beta/\alpha - 2(\beta + \gamma - 1) < 1.$$

Putting these three condition on Λ together implies that we must assume the following:

$$\Lambda < \min \left\{ d, 2\alpha, \frac{\alpha}{\beta} [2(\beta + \gamma) - 1] \right\}, \quad (4.5)$$

A:RieszPara

which is the same condition that was given in [CE22, Theorem 1.6] where the existence and uniqueness of (1.1) is proven but with a time-independent noise.

We now prove a lemma which is a generalization of equation (2.2) which accounts for the colored noise.

L:YfYbdd

Lemma 4.5. *Suppose Assumption 4.4 holds. Then there exists a constant $C(\beta, \gamma) =: C > 0$ such that for any $x, y \in \mathbb{R}^d$,*

$$\int_{\mathbb{R}^{2d}} Y(t, x - z) Y(t, y - w) f(w - z) \, dz dw \leq C \left(\frac{\nu}{2} \right)^{-\lambda/\alpha} t^{2(\beta+\gamma-1)-\beta\lambda/\alpha} \int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1 + |\xi|^\alpha)^2} < \infty \quad (4.6)$$

E:YYfBdd

Proof. Through a change of variables, we may write the left side of (4.6) as the following:

$$\int_{\mathbb{R}^{2d}} Y(t, x - z) Y(t, y - w) f(w - z) \, dz dw = \int_{\mathbb{R}^d} Y(t, u - (x - y)) (Y(t, \cdot) * f)(u) \, du,$$

where $*$ denotes the spatial convolution. Now an application of Parseval's identity gives us that

$$\int_{\mathbb{R}^d} Y(t, u - (x - y)) (Y(t, \cdot) * f)(u) \, du = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \mathcal{F}Y(t, \cdot - (x - y))(\zeta) \mathcal{F}(Y(t, \cdot) * f)(\zeta) d\zeta.$$

Applying the translation property of the Fourier transform and [CHN19, Theorem 4.1] gives us that

$$\begin{aligned} \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \mathcal{F}Y(t, \cdot - (x - y))(\zeta) \mathcal{F}(Y(t, \cdot) * f)(\zeta) d\zeta \\ = \frac{t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^d} \exp(-i(x - y)\zeta) E_{\beta, \beta+\gamma} \left(-\frac{\nu t^\beta |\zeta|^\alpha}{2} \right)^2 \mu(d\zeta). \end{aligned}$$

Hence by taking the absolute value on both sides and recalling the positivity of f , Y and the Mittag-Leffler function we get that

$$\int_{\mathbb{R}^{2d}} Y(t, x - z) Y(t, y - w) f(w - z) \, dz dw \leq \frac{t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^d} E_{\beta, \beta+\gamma} \left(-\frac{\nu t^\beta |\zeta|^\alpha}{2} \right)^2 \mu(d\zeta).$$

Now, by applying [Pod99, Theorem 1.6], we see that there exists a $C > 0$ depending only on β and γ such that

$$\frac{t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^d} E_{\beta, \beta+\gamma} \left(-\frac{\nu t^\beta |\xi|^\alpha}{2} \right)^2 \mu(d\xi) \leq \frac{C t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1 + \frac{\nu}{2} t^\beta |\zeta|^\alpha)^2}.$$

Another change of variables implies that

$$\frac{C t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1 + \frac{\nu}{2} t^\beta |\zeta|^\alpha)^2} \leq \frac{C}{(2\pi)^d} \left(\frac{\nu}{2} \right)^{-\lambda/\alpha} t^{2(\beta+\gamma-1)-\beta\lambda/\alpha} \int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1 + |\xi|^\alpha)^2}.$$

The finiteness of the above integral follows from Assumption 4.4. $\square$

4.1 Existence and uniqueness of a global solution with a globally Lipschitz drift term

We will now prove existence and uniqueness of the solution of (1.1) driven by a noise that is white in time and colored in space and when we take σ to be globally Lipschitz,

$$|\sigma(x) - \sigma(y)| \leq L(\sigma)|x - y|,$$

or if we take $c(\sigma) = \sigma(0)$,

$$|\sigma(x)| \leq c(\sigma) + L(\sigma)|x|.$$

Theorem 4.6. *Suppose that Assumption 4.4 holds, in particular the condition on Λ in (4.5), and that σ is globally Lipschitz. Then there exists a unique $L^2(\Omega)$ -solution to (1.1) driven by $\dot{F}$ such that*

$$u(t, x) = J_0(t, x) + I(t, x), \quad a.s.$$

Proof. The proof follows the strategy laid out in Walsh [Wal86]. We start by defining

$$u_0(t, x) := J_0(t, x) \quad \text{and} \quad u_n(t, x) := J_0(t, x) + \int_0^t \int_{\mathbb{R}^d} Y(t-s, x-y) \sigma(u_n(s, y)) F(ds, dy).$$

In addition, we define

$$D_n(t, x) := \mathbb{E} |u_{n+1}(t, x) - u_n(t, x)|^2 \quad \text{and} \quad H_n(t) := \sup_{x \in \mathbb{R}^d} D_n(t, x).$$

Lastly, we define

$$\Sigma(t, x, n) := |\sigma(u_n(t, x)) - \sigma(u_{n-1}(t, x))|.$$

With this notation we see that

$$D_n(t, x) = \int_0^t ds \int_{\mathbb{R}^d} dz dy Y(t-s, x-y) Y(t-s, x-z) f(y-z) \mathbb{E} (\Sigma(s, y, n) \Sigma(s, z, n)).$$

Now note that by the Lipschitz property of σ , we have that

$$\begin{aligned} \mathbb{E} (\Sigma(s, y, n) \Sigma(s, z, n)) &\leq L_\sigma^2 \mathbb{E} (|u_n(s, y) - u_{n-1}(s, y)| |u_n(s, z) - u_{n-1}(s, z)|) \\ &\leq L_\sigma^2 \|u_n(s, y) - u_{n-1}(s, y)\| \|u_n(s, z) - u_{n-1}(s, z)\| \\ &\leq L_\sigma^2 \sqrt{D_{n-1}(s, y) D_{n-1}(s, z)} \\ &\leq L_\sigma^2 H_{n-1}(s). \end{aligned}$$

Thus,

$$\begin{aligned} D_n(t, x) &\leq L_\sigma^2 \int_0^t H_{n-1}(s) \int_{\mathbb{R}^d} dz dy Y(t-s, x-y) Y(t-s, x-z) f(y-z) \\ &\leq CL_\sigma^2 \left(\frac{\nu}{2}\right)^{-\lambda/\alpha} \int_0^t ds \frac{H_{n-1}(s)}{(t-s)^{\lambda\beta/\alpha - 2(\beta+\gamma-1)}}, \end{aligned}$$

where the last inequality follows from Lemma 4.5. In return, this implies that

$$H_n(t) \leq CL_\sigma^2 \left(\frac{\nu}{2}\right)^{-\lambda/\alpha} \int_0^t ds \frac{H_{n-1}(s)}{(t-s)^{\lambda\beta/\alpha - 2(\beta+\gamma-1)}}.$$

Since we assume that $\lambda\beta/\alpha - 2(\beta+\gamma-1) < 1$, then the result follows from the discussion before the proof of [Wal86, Lemma 3.3]. $\square$

4.2 Some moment bounds for a globally Lipschitz drift term

We will now focus on proving some moment bounds that parallel those proved above in the space-time white noise case.

Proposition 4.7. [MS21, Proposition 4.4] *Let u_0 and v_0 be Borel functions satisfying $\|u_0\|_\infty + \|v_0\|_\infty < \infty$ and suppose that Assumption 4.4 holds. Also, in this proof we consider $\rho = \rho(\lambda)$ which is defined in (2.5). Then there exists a universal constant $K := K(\alpha, \beta, \gamma, d, \lambda) > 0$ such that for any*

$$\alpha^\rho > (8L^2(\sigma)K^2).$$

and

$$p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2}\right].$$

we have that

$$N_{a,p}(u) \leq 2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \quad (4.7) \quad \boxed{\text{E:ColorNbdd}}$$

where

$$\mathcal{T}_0 = \begin{cases} (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|u_0\|_\infty & \beta \in (0, 1] \\ (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases}$$

Moreover for any $T > 0$,

$$\sup_{x \in \mathbb{R}^d} \mathbb{E}(|u(t, x)|^p) \leq \exp(apt) \left[2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)}\right]^p, \quad t \in [0, T]. \quad (4.8) \quad \boxed{\text{E:ColorPMom}}$$

Proof. We note that $\mathcal{N}_{a,p}(J)$ has the same upper bound as in Proposition 2.4 above. In other words,

$$\mathcal{N}_{a,p}(J) \leq \begin{cases} \|u_0\|_\infty & \beta \in (0, 1] \\ (ea)^{-1} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases}$$

We know work to find an upper bound for $\mathcal{N}_{a,p}(I)$. The Burkholder-Davies-Gundy inequality and Hölder's inequality imply that

$$\|I(t, x)\|_p^2 \leq 4p \int_0^t ds \iint_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, x-y_1) f(x-y) Y(t-s, x-y_2) \|\sigma(u(s, y_1))\|_p \|\sigma(u(s, y_2))\|_p.$$

We now apply the Lipschitz property to see that for $i \in \{1, 2\}$,

$$\|\sigma(u(s, y_i))\|_p \leq c(\sigma) + L(\sigma) \|u(s, y_i)\|_p.$$

Therefore, we may say that

$$\begin{aligned} \|I(t, x)\|_p^2 &\leq 4p \int_0^t ds \iint_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, x-y_1) f(x-y) Y(t-s, x-y_2) \\ &\quad \times \prod_{i=1}^2 c(\sigma) + L(\sigma) \|u(s, y_i)\|_p \\ &\leq 4p \int_0^t ds \iint_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, x-y_1) f(x-y) Y(t-s, x-y_2) [c(\sigma) + L(\sigma) \exp(as) N_{a,p}(u)]^2 \\ &\leq 8pc(\sigma)^2 \int_0^t ds \iint_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, x-y_1) f(x-y) Y(t-s, x-y_2) \\ &\quad + 8pN_{a,p}(u)^2 L(\sigma)^2 \int_0^t ds \exp(2as) \iint_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, x-y_1) f(x-y) Y(t-s, x-y_2). \end{aligned}$$

Now by applying Lemma 4.5, we may bound the $\iint_{\mathbb{R}^{2d}} dy_1 dy_2$ integrals by the quantity

$$K_1(t-s)^{2(\beta+\gamma-1)-\beta\lambda/\alpha},$$

where

$$K_1 := CK_f \left(\frac{\nu}{2}\right)^{-\lambda/\alpha},$$

and C is as in Lemma 4.5. In return, this gives us that

$$\|I(t, x)\|_p^2 \leq 8pK_1 c(\sigma)^2 \int_0^t (t-s)^{2(\beta+\gamma-1)-\beta\lambda/\alpha} ds$$

$$\begin{aligned}
& + 8pK_1N_{a,p}(u)^2L(\sigma)^2 \int_0^t \exp(2as)(t-s)^{2(\beta+\gamma-1)-\beta\lambda/\alpha} ds \\
& = 8pK_1c(\sigma)^2 \frac{t^\rho}{\rho} + 8pK_1N_{a,p}(u)^2L(\sigma)^2 \frac{\exp(2at) [\Gamma(\rho) - \Gamma(\rho, 2at)]}{(2a)^\rho} \\
& =: \Delta_1(t) + \Delta_2(t),
\end{aligned}$$

where $\Gamma(\cdot)$ denotes the Gamma function and $\Gamma(\cdot, \circ)$ is the incomplete-upper Gamma function. Note that

$$\mathcal{N}_{a,p}(I)^2 \leq \sup_{t \geq 0} \exp(-2at) (\Delta_1(t) + \Delta_2(t)).$$

We start with bounding $\sup_{t \geq 0} \exp(-2at) \Delta_1(t)$. Recall the identities in (2.7). With this we see that

$$\sup_{t \geq 0} \exp(-2at) t^\rho = (\rho)^\rho (2ea)^{-\rho}.$$

Hence,

$$\sup_{t \geq 0} \exp(-2at) \Delta_1(t) \leq \frac{8pK_1c(\sigma)^2}{\rho} \left(\frac{\rho}{2ea} \right)^\rho.$$

Next we bound $\sup_{t \geq 0} \exp(-2at) \Delta_2(t)$. Note that for any $t \geq 0$,

$$\begin{aligned}
\exp(-2at) \Delta_2(t) & = 8pK_1N_{a,p}(u)^2L(\sigma)^2 \frac{[\Gamma(\rho) - \Gamma(\rho, 2at)]}{(2a)^\rho} \\
& \leq 8pK_1N_{a,p}(u)^2L(\sigma)^2 \frac{\Gamma(\rho)}{(2a)^\rho}.
\end{aligned}$$

Putting this all together gives us that

$$\mathcal{N}_{a,p}(I)^2 \leq \frac{8pK_1c(\sigma)^2}{\rho} \left(\frac{\rho}{2ea} \right)^\rho + \frac{8pK_1\Gamma(\rho)L(\sigma)^2}{(2a)^\rho} N_{a,p}(u)^2,$$

and thus,

$$\mathcal{N}_{a,p}(I) \leq 2c(\sigma) \sqrt{\frac{2pK_1}{\rho}} \left(\frac{\rho}{2ea} \right)^{\rho/2} + 2L(\sigma) \frac{\sqrt{2pK_1\Gamma(\rho)}}{(2a)^{\rho/2}} N_{a,p}(u).$$

Similar to the proof of Proposition 2.4 above, we define

$$K = K(\alpha, \beta, \gamma, d, \lambda) := \max \left\{ 2\sqrt{\frac{2K_1}{\rho}} \left(\frac{\rho}{2e} \right)^{\rho/2}, 2\frac{\sqrt{2K_1\Gamma(\rho)}}{(2)^{\rho/2}} \right\}.$$

We now see that

$$\mathcal{N}_{a,p}(I) \leq Kc(\sigma) \frac{\sqrt{p}}{a^{\rho/2}} + KL(\sigma) \frac{\sqrt{p}}{a^{\rho/2}} \mathcal{N}_{a,p}(u) \quad (4.9)$$

This and the bound for $\mathcal{N}_{a,p}(J)$ found above implies that

$$\begin{aligned}
\mathcal{N}_{a,p}(u) & \leq \mathcal{N}_{a,p}(J) + \mathcal{N}_{a,p}(I) \\
& \leq \begin{cases} \|u_0\|_\infty & \beta \in (0, 1] \\ (ea)^{-1} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2) \end{cases} \\
& \quad + Kc(\sigma) \frac{\sqrt{p}}{a^{\rho/2}} + KL(\sigma) \frac{\sqrt{p}}{a^{\rho/2}} \mathcal{N}_{a,p}(u).
\end{aligned} \quad (4.10)$$

Now consider

$$a^\rho > 8K^2L(\sigma)^2.$$

This implies that the interval

$$\left[2, \frac{a^\rho}{4K^2L(\sigma)^2}\right]$$

is nonempty. Hence, for any p in this interval we have that

$$\frac{KL(\sigma)\sqrt{p}}{a^{\rho/2}} \leq \frac{1}{2}.$$

This along with (4.10) implies that

$$\begin{aligned} \mathcal{N}_{a,p}(u) \leq & 2 \begin{cases} \|u_0\|_\infty & \beta \in (0, 1] \\ (ea)^{-1} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2) \end{cases} \\ & + \frac{c(\sigma)}{L(\sigma)}. \end{aligned}$$

Its easy to see that (4.8) follows directly from (4.7). $\square$

We now wish to prove a result similar to Proposition 2.8 above. More precisely, we wish to find a uniform bound of the p -th moment of the solution driven by a spatially colored noise and, in addition, we wish to prove the Hölder continuity of the stochastic integral.

Remark 4.8. Note that in [MS21], that they prove the Hölder continuity of the solution, not just the stochastic integral. What makes this more challenging in the case of the ISHWE is that calculating a meaningful upper bound on the $L^1(\mathbb{R}^d)$ increments of the fundamental solution is very challenging. However, as we saw in Proposition 2.5, we do have a meaningful upper bound for the $L^2(\mathbb{R}^d)$ increments of the fundamental solution Y .

However, due to the smoothness of Z and Z^* , the solution to the homogeneous equation, $J_0(t, x)$, is continuous and so proving the Hölder regularity of just the stochastic integral term will suffice. Nevertheless, the next conjecture gives some additional assumptions on the initial conditions, u_0 and v_0 , that may allow us to prove Hölder continuity of the solution itself rather than just the stochastic integral term.

Conjecture 4.9. Suppose that the initial data satisfy

$$u_0, v_0 \in L^1(\mathbb{R}^d) \cap L^\infty(\mathbb{R}^d).$$

Then we have the following:

$$|J_0(t, x) - J_0(s, y)| \leq \text{Const.} (|x - y|^{k_1} + |t - s|^{k_2}) \quad (4.11)$$

Proof. We will just consider the case where $\beta \in (1, 2)$ as this will include the case that $\beta \in (0, 1]$. As for the time increments, we see that

$$\begin{aligned} |([Z(t, \cdot) - Z(s, \cdot)] * v_0)(x)| &\leq \int_{\mathbb{R}^d} |Z(t, y) - Z(s, y)| |v_0(x - y)| dy \\ &\leq \sqrt{\int_{\mathbb{R}^d} |Z(t, y) - Z(s, y)|^2 |v_0(x - y)| dy} \sqrt{\int_{\mathbb{R}^d} |v_0(y)| dy} \\ &\leq \|v_0\|_{L^1(\mathbb{R}^d)}^{1/2} \|v_0\|_{L^\infty(\mathbb{R}^d)}^{1/2} \sqrt{\int_{\mathbb{R}^d} |Z(t, y) - Z(s, y)|^2 dy}. \end{aligned}$$

Similarly we can show that

$$|([Z^*(t, \cdot) - Z^*(s, \cdot)] * u_0)(x)| \leq \|u_0\|_{L^1(\mathbb{R}^d)}^{1/2} \|u_0\|_{L^\infty(\mathbb{R}^d)}^{1/2} \sqrt{\int_{\mathbb{R}^d} |Z^*(t, y) - Z^*(s, y)|^2 dy}.$$

Now if we consider the spatial increments, then by the same arguments as above we see that

$$\begin{aligned} |(Z(t, \cdot) * v_0)(x) - (Z(t, \cdot) * v_0)(y)| &\leq \int_{\mathbb{R}^d} |Z(t, x - z) - Z(t, y - z)| |v_0(z)| dz \\ &\leq \|v_0\|_{L^\infty(\mathbb{R}^d)}^{1/2} \|u_0\|_{L^1(\mathbb{R}^d)}^{1/2} \sqrt{\int_{\mathbb{R}^d} |Z(t, x - z) - Z(t, y - z)|^2 dz}, \end{aligned}$$

and in addition,

$$\begin{aligned} |(Z^*(t, \cdot) * v_0)(x) - (Z^*(t, \cdot) * u_0)(y)| &\leq \int_{\mathbb{R}^d} |Z^*(t, x - z) - Z^*(t, y - z)| |u_0(z)| dz \\ &\leq \|u_0\|_{L^\infty(\mathbb{R}^d)}^{1/2} \|u_0\|_{L^1(\mathbb{R}^d)}^{1/2} \sqrt{\int_{\mathbb{R}^d} |Z^*(t, x - z) - Z^*(t, y - z)|^2 dz}. \end{aligned}$$

Thus, under the assumptions of this conjecture, it will suffice to calculate the $L^2(\mathbb{R}^d)$ increments in space and time of Z and Z^* which should not be a problem. $\square$

We now state a lemma which will be needed in the proof of the next proposition.

phiParseval

Lemma 4.10. *Let ϕ and ψ be bounded Borel functions. Then for $s, t > 0$ and $z \in \mathbb{R}^d$ we have,*

$$\iint_{\mathbb{R}^{2d}} dx dy \phi(x) Y(t, x) \psi(y) Y(s, y) f(x - y + z) = (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F}(\phi Y(t, \cdot))(\xi) \overline{\mathcal{F}(\psi Y(s, \cdot))(\xi)} e^{-iz \cdot \xi} \mu(d\xi) \quad (4.12)$$

Proof. For notation, define $Y_\phi(t, x) = \phi(x) Y(t, x)$ and $Y_\psi(t, x) = \psi(x) Y(t, x)$. The boundedness and measurability of ϕ and ψ guarantees the existence of $\mathcal{F}Y_\phi$ and $\mathcal{F}Y_\psi$. Thus we have,

$$\iint_{\mathbb{R}^{2d}} dx dy Y_\phi(t, x) Y_\psi(s, y) f(x - y + z) = \int_{\mathbb{R}^d} dx Y_\phi(t, x) (Y_\psi(s, \cdot) * f(\cdot + z))(x).$$

Parseval's identity implies that

$$\begin{aligned} \int_{\mathbb{R}^d} dx Y_\phi(t, x) (Y_\psi(s, \cdot) * f(\cdot + z))(x) &= (2\pi)^{-d} \int_{\mathbb{R}^d} d\zeta \mathcal{F}(Y_\phi(t, \cdot))(\zeta) \overline{\mathcal{F}((Y_\psi(s, \cdot))(\zeta) \mathcal{F}(f(\cdot + z)))} \\ &= (2\pi)^{-d} \int_{\mathbb{R}^d} \mu(d\zeta) \mathcal{F}(Y_\phi(t, \cdot))(\zeta) \overline{\mathcal{F}((Y_\psi(s, \cdot))(\zeta) e^{-iz \cdot \zeta})}, \end{aligned}$$

where $\mu(d\xi) = \varphi(\xi) d\xi$. This completes the proof. $\square$

A:stronger

Assumption 4.11. The following proposition will require us to again strengthen our assumption on the Riesz parameter Λ . We now require that for some $0 < r < 1$, in addition to the assumptions in (4.5), that

$$2(\beta + \gamma - 1) - \frac{\beta\lambda}{\alpha} > 0 \quad \text{and} \quad 2(\beta + \gamma - 1) - \frac{\beta\lambda}{\alpha} - \frac{2r\beta}{\alpha} > -1$$

or in other words,

$$\Lambda < \frac{\alpha}{\beta} [2(\beta + \gamma) - 2] \quad \text{and} \quad \Lambda < \frac{\alpha}{\beta} [2(\beta + \gamma) - 1] - 2r.$$

Hence the condition on Λ becomes

$$0 < \Lambda < \min\{d, 2\alpha, \frac{\alpha}{\beta} [2(\beta + \gamma) - 2], \frac{\alpha}{\beta} [2(\beta + \gamma) - 1] - 2r\}. \quad (4.13)$$

A:RieszPara

Remark 4.12. Note that the additional assumption (4.13) excludes the SHE, $\alpha = 2$, $\beta = 1$ and $\gamma = 1$, as well as the popular case where $\gamma = 1 - \beta$.

Proposition 4.13. [MS21, Proposition 4.8] Let $T > 0$. Under Assumption 4.11, we have that for any $x, z \in \mathbb{R}^d$ and $0 \leq t \leq T$,

$$\begin{aligned} & \|I(t, x) - I(t, z)\|_p^2 \\ & \leq C|x - z|^{2r} \left[\int_{\mathbb{R}^d} \mu(d\zeta) \frac{|\zeta|^{2r}}{(1 + |\zeta|^\alpha)^2} \right] \left[\int_0^t ds (t - s)^{2(\beta + \gamma - 1) - \frac{\beta\lambda}{\alpha} - \frac{2r\beta}{\alpha}} \right] \left[c(\sigma) + L(\sigma) \sup_{(s, y) \in [0, T] \times \mathbb{R}^d} \|u(s, y)\|_p \right]^2 \\ & \quad + C \left[\int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1 + |\zeta|^\alpha)^2} \right] L^2(\sigma) t^{2(\beta + \gamma - 1) - \beta\lambda/\alpha} \int_0^t ds \sup_{|z_1 - z_2| = |\xi|} \|u(s, z_1) - u(s, z_2)\|_p^2. \end{aligned}$$

Proof. The following proof follows closely the techniques presented in [HHN14, Theorem 3.1]. By the Burkholder Davies Gundy inequality, we have that

$$\begin{aligned} \|I(t, x) - I(t, z)\|_p^2 & \leq 4p \left\| \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 f(y_1 - y_2) \prod_{i=1}^2 (Y(t - s, x - y_i) - Y(t - s, z - y_i) \sigma(u(s, y_i))) \right\|_{p/2}^2 \\ & =: 4p \|Q\|_{p/2} \end{aligned}$$

We now reorganize the integrand of Q above by following [HHN14, Theorem 3.1]. By expanding the integrand of Q , we get that

$$\begin{aligned} Q &= \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t - s, x - y_1) Y(t - s, x - y_2) f(y_1 - y_2) \sigma(u(s, y_1)) \sigma(u(s, y_2)) \\ & \quad - \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t - s, x - y_1) Y(t - s, z - y_2) f(y_1 - y_2) \sigma(u(s, y_1)) \sigma(u(s, y_2)) \\ & \quad - \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t - s, z - y_1) Y(t - s, x - y_2) f(y_1 - y_2) \sigma(u(s, y_1)) \sigma(u(s, y_2)) \\ & \quad + \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t - s, z - y_1) Y(t - s, z - y_2) f(y_1 - y_2) \sigma(u(s, y_1)) \sigma(u(s, y_2)) \\ & =: I_1 + I_2 + I_3 + I_4. \end{aligned}$$

Now let $\xi = x - z$ and consider the following notation:

$$\begin{aligned} \Sigma_x(s, y) &= \sigma(u(s, x - y)), \quad \Sigma_{x,z}(s, y) = \sigma(u(s, x - y)) - \sigma(u(s, z - y)), \\ h_1(s, y_1, y_2) &= f(y_1 - y_2) \Sigma_{x,z}(s, y_1) \Sigma_{x,z}(s, y_2) \\ h_2(s, y_1, y_2) &= [f(y_1 - y_2 + \xi) - f(y_1 - y_2)] \Sigma_x(s, y_2) \Sigma_{x,z}(s, y_1) \\ h_3(s, y_1, y_2) &= h_2(s, y_2, y_1) \\ h_4(s, y_1, y_2) &= [2f(y_1 - y_2) - f(y_1 - y_2 + \xi) - f(y_1 - y_2 - \xi)] \Sigma_x(s, y) \Sigma_x(s, z). \end{aligned}$$

With this notation along with a change of variables, we rewrite I_1 through I_4 as the following:

$$\begin{aligned} I_1 &= \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t - s, y_1) Y(t - s, y_2) f(y_1 - y_2) \Sigma_x(s, y_1) \Sigma_x(s, y_2) \\ I_2 &= - \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t - s, y_1) Y(t - s, y_2) f(y_1 - y_2 - \xi) \Sigma_x(s, y_1) \Sigma_z(s, y_2) \\ &= \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t - s, y_1) Y(t - s, y_2) f(y_1 - y_2 - \xi) \Sigma_x(s, y_1) \Sigma_{x,z}(s, y_2) \\ & \quad - \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t - s, y_1) Y(t - s, y_2) f(y_1 - y_2 - \xi) \Sigma_x(s, y_1) \Sigma_x(s, y_2) \end{aligned}$$

$$\begin{aligned}
I_3 &= \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, y_1) Y(t-s, y_2) f(y_1 - y_2 + \xi) \Sigma_x(s, y_2) \Sigma_{x,z}(s, y_1) \\
&\quad - \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, y_1) Y(t-s, y_2) f(y_1 - y_2 + \xi) \Sigma_x(s, y_1) \Sigma_x(s, y_2) \\
I_4 &= \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, y_1) Y(t-s, y_2) f(y_1 - y_2) \Sigma_z(s, y_1) \Sigma_z(s, y_2) \\
&= \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, y_1) Y(t-s, y_2) f(y_1 - y_2) (\Sigma_{x,z}(s, y_1) - \Sigma_x(s, y_1) (\Sigma_{x,z}(s, y_2) - \Sigma_x(s, y_2))),
\end{aligned}$$

where above I_3 is derived in the same way I_2 was derived. Now if we expand I_4 and regroup all the terms in $I_1 + \dots + I_4$, we get that

$$Q = \sum_{i=1}^4 Q_i \quad (4.14)$$

where

$$Q_i = \int_0^t \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, y_1) Y(t-s, y_2) h_i(s, y_1, y_2). \quad (4.15)$$

Thus

$$\|I(t, x) - I(t, z)\|_p^2 \leq 4p \sum_{i=1}^4 \|Q_i\|_{p/2}. \quad (4.16)$$

E: Ispaceinc

We now estimate each $\|Q_i\|_{p/2}$ separately.

Upper bound for $\|Q_1\|_{p/2}$. Using the Lipschitz property of σ , Minkowski's inequality and the triangle inequality gives is

$$\begin{aligned}
\|Q_1\|_{p/2} &\leq L(\sigma)^2 \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, y_1) Y(t-s, y_2) f(y_1 - y_2) \left[\sup_{|z_1 - z_2| = \xi} \|u(s, z_1) - u(s, z_2)\|_p^2 \right] \\
&\leq L(\sigma)^2 C K_f \left(\frac{\nu}{2}\right)^{-\lambda/\alpha} \int_0^t ds (t-s)^{2(\beta+\gamma-1)-\beta\lambda/\alpha} \left[\sup_{|z_1 - z_2| = \xi} \|u(s, z_1) - u(s, z_2)\|_p^2 \right] \\
&\leq L(\sigma)^2 C K_f \left(\frac{\nu}{2}\right)^{-\lambda/\alpha} (t)^{2(\beta+\gamma-1)-\beta\lambda/\alpha} \int_0^t ds \left[\sup_{|z_1 - z_2| = \xi} \|u(s, z_1) - u(s, z_2)\|_p^2 \right],
\end{aligned}$$

where the second inequality above follows from Lemma 4.5 and the last line requires $2(\beta + \gamma - 1) - \beta\lambda/\alpha > 0$.

To bound the remaining Q'_i s, we will need to consider the following truncations which will then allow us to apply Lemma 4.10. For $k \geq 1$ we define $\Sigma_x^k(s, y) = \Sigma_x(s, y) 1_{\{\Sigma_x(s, y) \leq k\}}$ and $\Sigma_{x,z}^k(s, y) = \Sigma_{x,z}(s, y) 1_{\{\Sigma_{x,z}(s, y) \leq k\}}$ and then we define Q_i^k the same as Q_i but with h_i replaced with h_i^k where h_i^k is defined by replacing $\Sigma_x(s, y)$ and $\Sigma_{x,z}(s, y)$ with $\Sigma_x^k(s, y)$ and $\Sigma_{x,z}^k(s, y)$ respectively in h_i .

Upper Bound for $\|Q_2^k\|_{p/2}$. We apply Lemma 4.10 with $\phi(y_2) = \Sigma_x^k(s, y_2)$ and $\psi(y_1) = \Sigma_{x,z}^k(s, y_1)$. Thus we have that up to a constant of $(2\pi)^{-d}$,

$$Q_2^k = \int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) \overline{\mathcal{F}(\Sigma_x^k(s, \cdot) Y(t-s, \cdot))}(\zeta) \mathcal{F}(\Sigma_{x,z}^k Y(t-s, \cdot))(\zeta) [e^{-i\xi \cdot \zeta} - 1],$$

where we recall that $\xi = x - z$. We now apply the the following to bounds: $|e^{-i\xi \cdot \zeta} - 1| \leq C|\xi|^r |\zeta|^r$ for any $r \in [0, 1]$ and some $C > 0$ and $\sqrt{ab} \leq \frac{1}{2}(a+b)$ for and $a, b > 0$. Indeed, for some $r \in [0, 1]$,

$$|Q_2^k| \leq C \int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) \left| \mathcal{F}(\Sigma_x^k(s, \cdot) Y(t-s, \cdot))(\zeta) \right| \left| \mathcal{F}(\Sigma_{x,z}^k Y(t-s, \cdot))(\zeta) \right| |\xi|^r |\zeta|^r$$

$$\begin{aligned}
&\leq \int_0^t ds |\xi|^r \left(\int_{\mathbb{R}^d} \mu(d\zeta) \left| \mathcal{F} \left(\Sigma_x^k(s, \cdot) Y(t-s, \cdot) \right) (\zeta) \right|^2 |\zeta|^{2r} \right)^{1/2} \\
&\quad \times \left(\int_{\mathbb{R}^d} \mu(d\zeta) \left| \mathcal{F} \left(\Sigma_{x,z}^k Y(t-s, \cdot) \right) (\zeta) \right|^2 \right)^{1/2} \\
&\leq \frac{1}{2} \int_0^t ds |\xi|^{2r} \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} \left| \mathcal{F} \left(\Sigma_x^k(s, \cdot) Y(t-s, \cdot) \right) (\zeta) \right|^2 \\
&\quad + \int_{\mathbb{R}^d} \mu(d\zeta) \left| \mathcal{F} \left(\Sigma_{x,z}^k(s, \cdot) Y(t-s, \cdot) \right) (\zeta) \right|^2 \\
&=: Q_{2,1}^k + Q_{2,2}^k.
\end{aligned}$$

As for $Q_{2,1}^k$ ¹,

$$\begin{aligned}
\|Q_{2,1}^k\|_{p/2} &\leq C |\xi|^{2r} \left\| \sup_{(s,y) \in [0,T] \times \mathbb{R}^d} \Sigma_{x,z}^k(s, y) \int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} |\mathcal{F}(Y(t-s, \cdot))(\zeta)|^2 \right\|_{p/2} \\
&\leq C |\xi|^{2r} \sup_{(s,y) \in [0,T] \times \mathbb{R}^d} \left\| \Sigma_{x,z}^k(s, y) \right\|_p^2 \int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} |\mathcal{F}(Y(t-s, \cdot))(\zeta)|^2 \\
&\leq C |\xi|^{2r} \left[c(\sigma) + L(\sigma) \sup_{(s,y) \in [0,T] \times \mathbb{R}^d} \|u(s, y)\|_p \right]^2 \int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} |\mathcal{F}(Y(t-s, \cdot))(\zeta)|^2
\end{aligned}$$

which follows from the global Lipschitz assumption on σ . Now as for $Q_{2,2}^k$,

$$\begin{aligned}
\|Q_{2,2}^k\|_{p/2} &\leq \int_0^t ds \sup_{y \in \mathbb{R}^d} \left\| \Sigma_{x,z}^k(s, y) \right\|_p^2 \int_{\mathbb{R}^d} \mu(d\zeta) |\mathcal{F}(Y(t-s, \cdot))(\zeta)|^2 \\
&\leq C \int_0^t ds \sup_{y \in \mathbb{R}^d} \left\| \Sigma_{x,z}^k(s, y) \right\|_p^2 (t-s)^{2(\beta+\gamma-1)-\beta\lambda/\alpha} \int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1+|\zeta|^\alpha)^2} \\
&\leq C \left(\int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1+|\zeta|^\alpha)^2} \right) L^2(\sigma) \int_0^t ds (t-s)^{2(\beta+\gamma-1)-\beta\lambda/\alpha} \sup_{|z_1-z_2|=|\xi|} \|u(s, z_1) - u(s, z_2)\|_p^2 \\
&\leq C \left(\int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1+|\zeta|^\alpha)^2} \right) L^2(\sigma) t^{2(\beta+\gamma-1)-\beta\lambda/\alpha} \int_0^t ds \sup_{|z_1-z_2|=|\xi|} \|u(s, z_1) - u(s, z_2)\|_p^2,
\end{aligned}$$

where in the second to last line we applied Lemma 4.5 and in the last line we assume that $2(\beta+\gamma-1)-\beta\lambda/\alpha \geq 0$. Summarizing the above gives us that

$$\begin{aligned}
\|Q_2^k\|_{p/2} &\leq C |\xi|^{2r} \left[c(\sigma) + L(\sigma) \sup_{(s,y) \in [0,T] \times \mathbb{R}^d} \|u(s, y)\|_p \right]^2 \int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} |\mathcal{F}(Y(t-s, \cdot))(\zeta)|^2 \\
&\quad + C \left(\int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1+|\zeta|^\alpha)^2} \right) L^2(\sigma) t^{2(\beta+\gamma-1)-\beta\lambda/\alpha} \int_0^t ds \sup_{|z_1-z_2|=|\xi|} \|u(s, z_1) - u(s, z_2)\|_p^2.
\end{aligned}$$

Also by swapping x and z and noticing that $|e^{-i\xi\zeta} - 1| = |e^{i\xi\zeta} - 1|$ then the bound for $\|Q_3^k\|_{p/2}$ with be the same as for $\|Q_2^k\|_{p/2}$.

Lastly, for Q_4^k we have that after applying Lemma 4.10 with $\phi = \psi = \Sigma_x^k(s, \cdot)$,

$$|Q_4^k| \leq C \int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) \left| 2 - e^{-i\xi\zeta} - e^{i\xi\zeta} \right| \left| \mathcal{F} \left(\Sigma_x^k(s, \cdot) Y(t-s, \cdot) \right) (\zeta) \right|^2$$

¹There is a comment made in [MS21, Proposition 4.8] and [HHN14, Theorem 3.2] that minkowski inequality is being used for the first inequality but it seems incorrect how they apply minkowski's and also unnecessary.

$$\begin{aligned}
&\leq C|\xi|^{2r} \int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} \left| \mathcal{F} \left(\Sigma_x^k(s, \cdot) Y(t-s, \cdot) \right) (\zeta) \right|^2 \\
&= C|\xi|^{2r} \int_0^t ds \int_{\mathbb{R}^d} dy g(y) \left[\left(\Sigma_x^k(s, \cdot) Y(t-s, \cdot) \right) * \left(\widetilde{\Sigma_x^k(s, \cdot) Y(t-s, \cdot)} \right) \right] (y),
\end{aligned}$$

where $g = \mathcal{F}^{-1} (\mu(\cdot) |\cdot|^{2r})$. Thus we see that

$$\begin{aligned}
\|Q_4^k\|_{p/2} &\leq C|\xi|^{2r} \int_0^t ds \sup_{y \in \mathbb{R}^d} \left\| \Sigma_x^k(s, y) \right\|_p^2 \int_{\mathbb{R}^d} dy g(y) \left[(Y(t-s, \cdot)) * (\widetilde{Y(t-s, \cdot)}) \right] (y) \\
&= C|\xi|^{2r} \sup_{(s,y) \in [0,T] \times \mathbb{R}^d} \left\| \Sigma_x^k(s, y) \right\|_p^2 \int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} |\mathcal{F}(Y(t-s, \cdot))(\zeta)|^2 \\
&\leq C|\xi|^{2r} \left[c(\sigma) + L(\sigma) \sup_{(s,y) \in [0,T] \times \mathbb{R}^d} \|u(s, y)\|_p \right]^2 \int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} |\mathcal{F}(Y(t-s, \cdot))(\zeta)|^2.
\end{aligned}$$

Thus taking (4.16) into consideration, we have that

$$\begin{aligned}
\|I(t, x) - I(t, z)\|_p^2 &\leq C|\xi|^{2r} \left[c(\sigma) + L(\sigma) \sup_{(s,y) \in [0,T] \times \mathbb{R}^d} \|u(s, y)\|_p \right]^2 \int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} |\mathcal{F}(Y(t-s, \cdot))(\zeta)|^2 \\
&\quad + C \left(\int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1 + |\zeta|^\alpha)^2} \right) L^2(\sigma) t^{2(\beta+\gamma-1)-\beta\lambda/\alpha} \int_0^t ds \sup_{|z_1-z_2|=|\xi|} \|u(s, z_1) - u(s, z_2)\|_p^2.
\end{aligned}$$

Lastly we note that by the same techniques in Lemma 4.5, we see that

$$\begin{aligned}
&\int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} |\mathcal{F}(Y(t-s, \cdot))(\zeta)|^2 \\
&\leq \int_0^t ds (t-s)^{2(\beta+\gamma-1)} \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} E_{\beta, \beta+\gamma} \left(-2^{-1} \nu(t-s)^\beta |\zeta|^\alpha \right)^2 \\
&\leq C \int_0^t ds (t-s)^{2(\beta+\gamma-1)} \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} \left(1 + 2^{-1} \nu(t-s)^\beta |\zeta|^\alpha \right)^{-2} \\
&= C \int_0^t ds (t-s)^{2(\beta+\gamma-1) - \frac{\beta\lambda}{\alpha} - \frac{2r\beta}{\alpha}} \int_{\mathbb{R}^d} \mu(d\zeta) \frac{|\zeta|^{2r}}{(1 + |\zeta|^\alpha)^2},
\end{aligned}$$

therefore,

$$\begin{aligned}
&\|I(t, x) - I(t, z)\|_p^2 \\
&\leq C|\xi|^{2r} \left[\int_{\mathbb{R}^d} \mu(d\zeta) \frac{|\zeta|^{2r}}{(1 + |\zeta|^\alpha)^2} \right] \left[\int_0^t ds (t-s)^{2(\beta+\gamma-1) - \frac{\beta\lambda}{\alpha} - \frac{2r\beta}{\alpha}} \right] \left[c(\sigma) + L(\sigma) \sup_{(s,y) \in [0,T] \times \mathbb{R}^d} \|u(s, y)\|_p \right]^2 \\
&\quad + C \left[\int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1 + |\zeta|^\alpha)^2} \right] L^2(\sigma) t^{2(\beta+\gamma-1) - \beta\lambda/\alpha} \int_0^t ds \sup_{|z_1-z_2|=|\xi|} \|u(s, z_1) - u(s, z_2)\|_p^2,
\end{aligned}$$

which requires us to assume that

$$2(\beta + \gamma - 1) - \frac{\beta\lambda}{\alpha} > 0 \quad \text{and} \quad 2(\beta + \gamma - 1) - \frac{\beta\lambda}{\alpha} - \frac{2r\beta}{\alpha} > -1$$

□

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